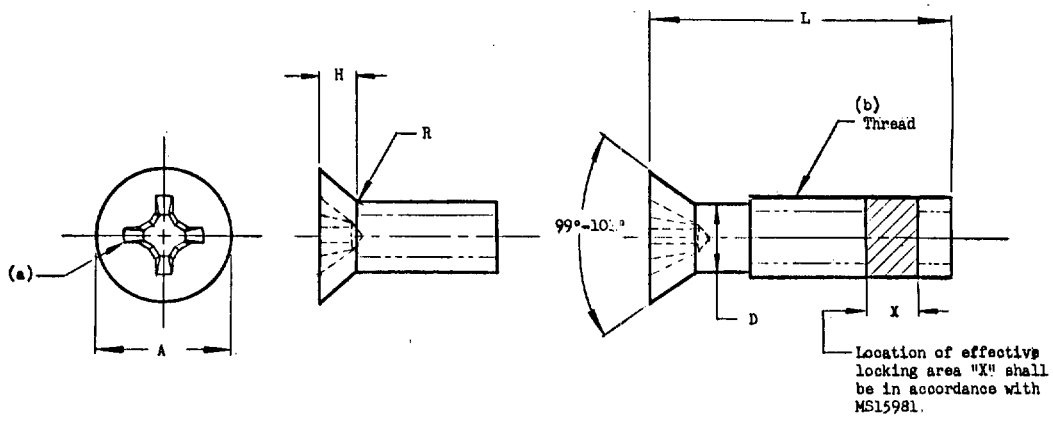
USER SYMBOLS:
ARMY - ME

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USAF - 99
DLA - IS

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| DASH NO DIA | THREAD (b) | D DIA | | A | | | H | | R | ULTIMATE TENSILE STRENGTH LBS MIN | DRIVER SIZE |
|---|---|---|---|---|---|---|---|---|---|---|---|
| | | MAX | MIN | SHARP MAX | SHARP MIN | ABS MIN | MAX | MIN | MAX | | |
| 02 | 2-56UNC-2A | .0860 | .0717 | .182 | .156 | .150 | .043 | .032 | .036 | .200 | 1 |
| 04 | 4-40UNC-2A | .1120 | .0925 | .225 | .207 | .200 | .048 | .040 | .050 | .380 | |
| 06 | 6-32UNC-2A | .1380 | .1141 | .279 | .257 | .249 | .060 | .051 | .062 | .545 | 2 |
| 08 | 8-32UNC-2A | .1640 | .1399 | .332 | .308 | .300 | .072 | .062 | | .770 | |
| 3 | 10-32UNF-2A | .1900 | .1658 | .385 | .357 | .348 | .083 | .072 | | 1,000 | |
| 4 | 1/4-28UNF-2A | .2500 | .2225 | .507 | .477 | .462 | .110 | .097 | .071 | 2,000 | 3 |
| 5 | 5/16-24UNF-2A | .3125 | .2806 | .635 | .600 | .581 | .138 | .123 | .083 | 3,200 | 4 |
| 6 | 3/8-24UNF-2A | .3750 | .3430 | .762 | .722 | .700 | .165 | .148 | | 4,830 | |

(a) DRIVE RECESS: Recess shall be in accordance with MS9006.

(b) Threads shall be in accordance with Screw Threads Standards for Federal Services, Handbook H-28. The threads shall extend to within 2 threads or less of head surface or screw length 2 inches or shorter. Lengths over 2 inches shall have a minimum thread length of 1-3/4 inches.

(c) Based on stress area equal to $\pi\left(\frac{E}{2} - \frac{3H}{16}\right)^2$ See Handbook H-28.

MATERIAL: Carbon Steel, FED-STD-66

(B) PLATING: Cadmium Plated, Specification QQ-P-416, Type II, Class 2.

LOCKING ELEMENT: In accordance with MIL-F-18240. Locking element area must pass through a ring gage equal to maximum major thread diameter +.010" freely with finger pressure.

DESIGN AND USAGE LIMITATIONS: See MS15981

EXAMPLE OF PART NUMBER: MS21093-0201=Screw, Self Locking 250° F, Carbon Steel, Cadmium plated, 2-56UNC-2A, 3/16 long.
(B) MS21093-343=Screw, Self Locking 250° F, Carbon Steel, Cadmium plated, 10-32UNF-2A, 3/8 long.
Unassigned Dash Numbers shall not be used.

DIMENSIONS: All dimensions are in inches.

Reference documents shall be of the issue in effect on date of invitations for bid. This standard takes precedence over documents referenced herein.

(B) DENOTES CHANGES.

APPROVED 2 Nov 65 REVISED (A) 11 Sept 67 (B) 30 Sept 77

| | | |
|---|---|---|
| P.A. NAVY- AS<br>Other Cust<br>USAF- 11<br>ARMY - AV | TITLE<br>SCREW, SELF-LOCKING, 250° F, STEEL, 55 KSI Ftu, 100° FLAT HEAD, CROSS RECESSED | MILITARY STANDARD<br>MS21093 |
| PROCUREMENT SPECIFICATION<br>MIL-F-18240/FF-S-92 | SUPERSEDES: | SHEET 1 OF 2 |

DD FORM 1 MAR 72 672-1 (Coordinated) Preparing Activity: Navy - AS PROJECT NO. 5305-1342

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| LENGTH DASH NUMBER | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| L | | THREAD SIZE AND PITCH | | | | | | | |
| LENGTH | TOLERANCE | 2-56 | 4-40 | 6-32 | 8-32 | 10-32 | 1/4-28 | 5/16-24 | 3/8-24 |
| (B) .1875 | (a) | 01 | 89 | | | | | | |
| .250 | | 02 | 09 | 18 | | | | | |
| .312 | | 03 | 10 | 19 | 31 | | | | |
| .375 | | 04 | 11 | 20 | 32 | 43 | | | |
| .4375 | +.000 -.031 | 05 | 12 | 21 | 33 | 44 | | | |
| .500 | | 06 | 13 | 22 | 34 | 45 | 56 | | |
| .625 | | 07 | 14 | 23 | 35 | 46 | 57 | 67 | 77 (a) |
| .750 | | 08 | 15 | 24 | 36 | 47 | 58 | 68 | 78 |
| .875 | | | 16 | 25 | 37 | 48 | 59 | 69 | 79 |
| 1.000 | | | 17 | 26 | 38 | 49 | 60 | 70 | 80 |
| 1.250 | | | | 27 | 39 | 50 | 61 | 71 | 81 |
| 1.500 | +.000 -.062 | | | 28 | 40 | 51 | 62 | 72 | 82 |
| 1.750 | | | | 29 | 41 | 52 | 63 | 73 | 83 |
| 2.000 | | | | 30 | 42 | 53 | 64 | 74 | 84 |
| 2.250 | | | | | | 54 | 65 | 75 | 85 |
| 2.500 | +.000 -.094 | | | | | 55 | 66 | 76 | 86 |
| 2.750 | | | | | | | | | 87 |
| 3.000 | | | | | | | | | 88 |

(a) MIL-F-18240 does not establish requirements for length of threads less than 9 pitches. Therefore, only positive indication of torque is required for screws having lengths above this heavy line.

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APPROVED 2 Nov 65
REVISED (B) FOR CHANGE SEE SHEET 1

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| P.A. NAVY - AS<br>Other Cust<br>USAF - 11<br>ARMY - AV | TITLE<br>SCREW, SELF-LOCKING, 250° F, STEEL, 55 KSI<br>Ftu, 100° FLAT HEAD, CROSS RECESSED | MILITARY STANDARD<br>MS21093 |
|---|---|---|
| PROCUREMENT SPECIFICATION<br>MIL-F-18240/FF-S-92 | SUPERSEDES: | SHEET 2 OF |



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